

Automated Data Analysis & Machine Learning for Ion-Selective Electrode Calibration

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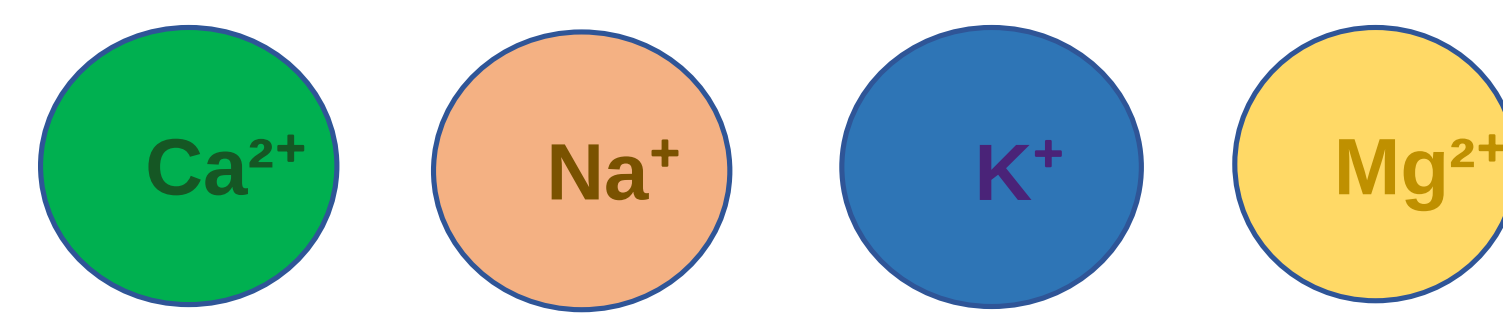
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1. Motivation

1. Water Quality Assessment



2. Concentration of Dissolved Ions in Water



3. Solid-Contact Ion-Selective Electrode (SC-ISE)

- Low cost
- Reliable
- Continuous real-time monitoring



4. Calibration Needed

Each electrode must be calibrated to convert potential into concentration using:

- Nernstian slope
- Intercept
- Linear range
- Limit of detection

5. Manual Calibration (Problem)

- Linear region chosen by eye
- Drift and failures easy to overlook
- Results vary from user to user
- With many electrodes over months, data accumulates quickly

6. Automated Calibration with Python

7. Why It Matters

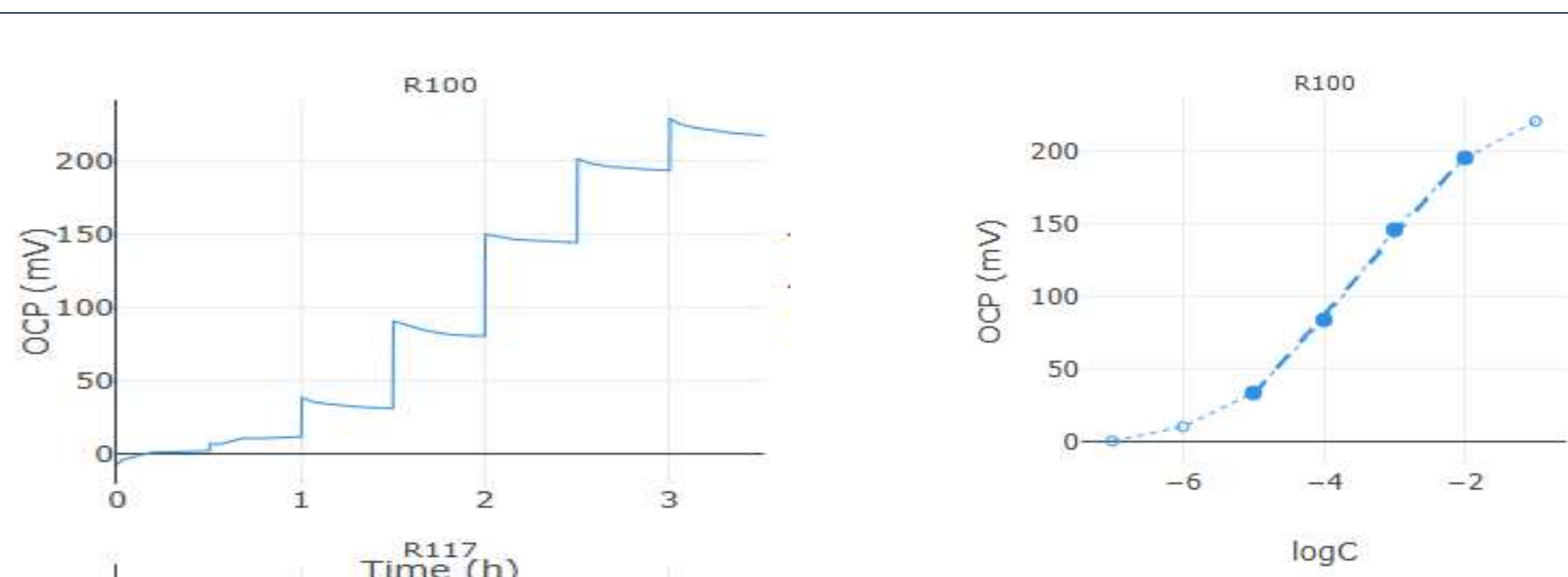
- Turns hours of repetitive spreadsheet work into seconds.
- Results are objective and reproducible across users.
- Cast calibration as a data problem, opening the door to ML-driven analysis and smarter electrode diagnostics.

2. Approach: Analysis Pipeline

- Ingest raw datasets → Clean & standardize → Plot potential vs. time for every electrode
- Auto-detect concentration-change steps and extract the steady-state OCP (average or last) at each concentration.
- Build the calibration curve (OCP vs. logC) for each electrode.
- Auto-detect the linear (Nernstian) region and fit it → store slope, intercept, R² score and compute cross-sensor correlation.
- Track behavior of sensor over time by reorganizing date-wise data → sensor-wise data.
- Deliver the whole pipeline as an interactive Streamlit web app.

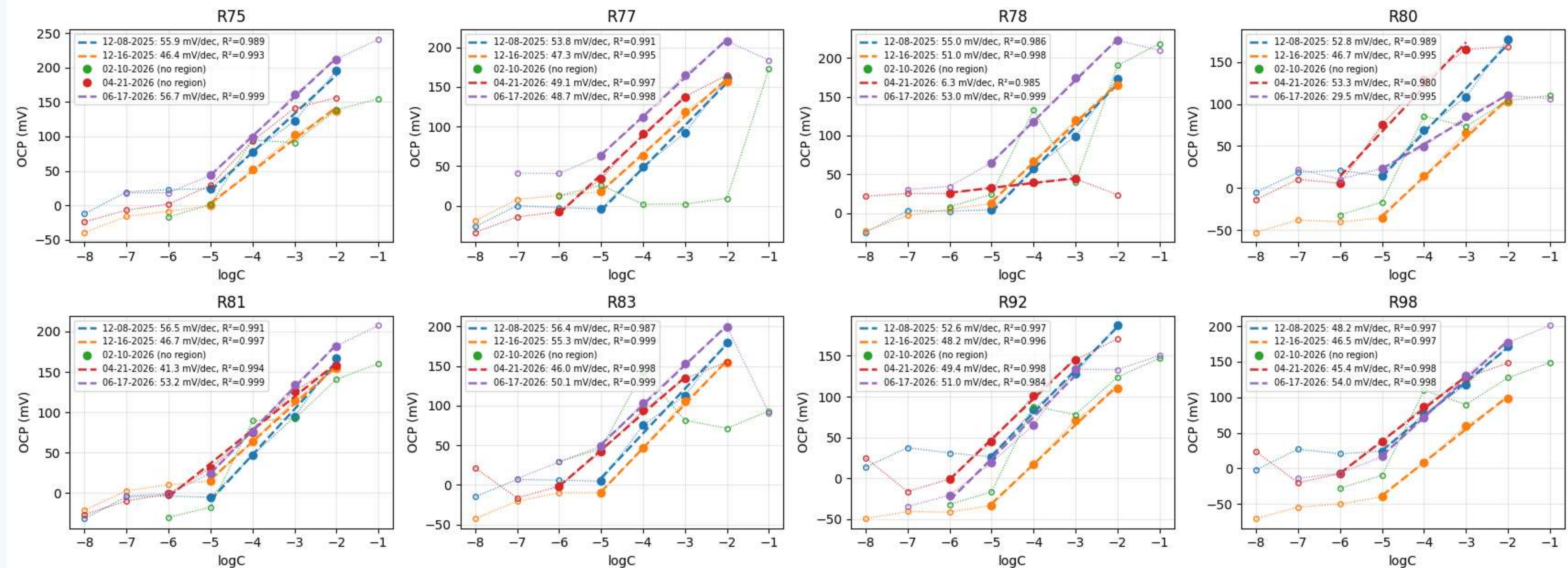
Implemented in Python (pandas, NumPy, Plotly, SciPy); deployed with Streamlit.

3. Signal Cleaning & Automatic Linear-Region Selection



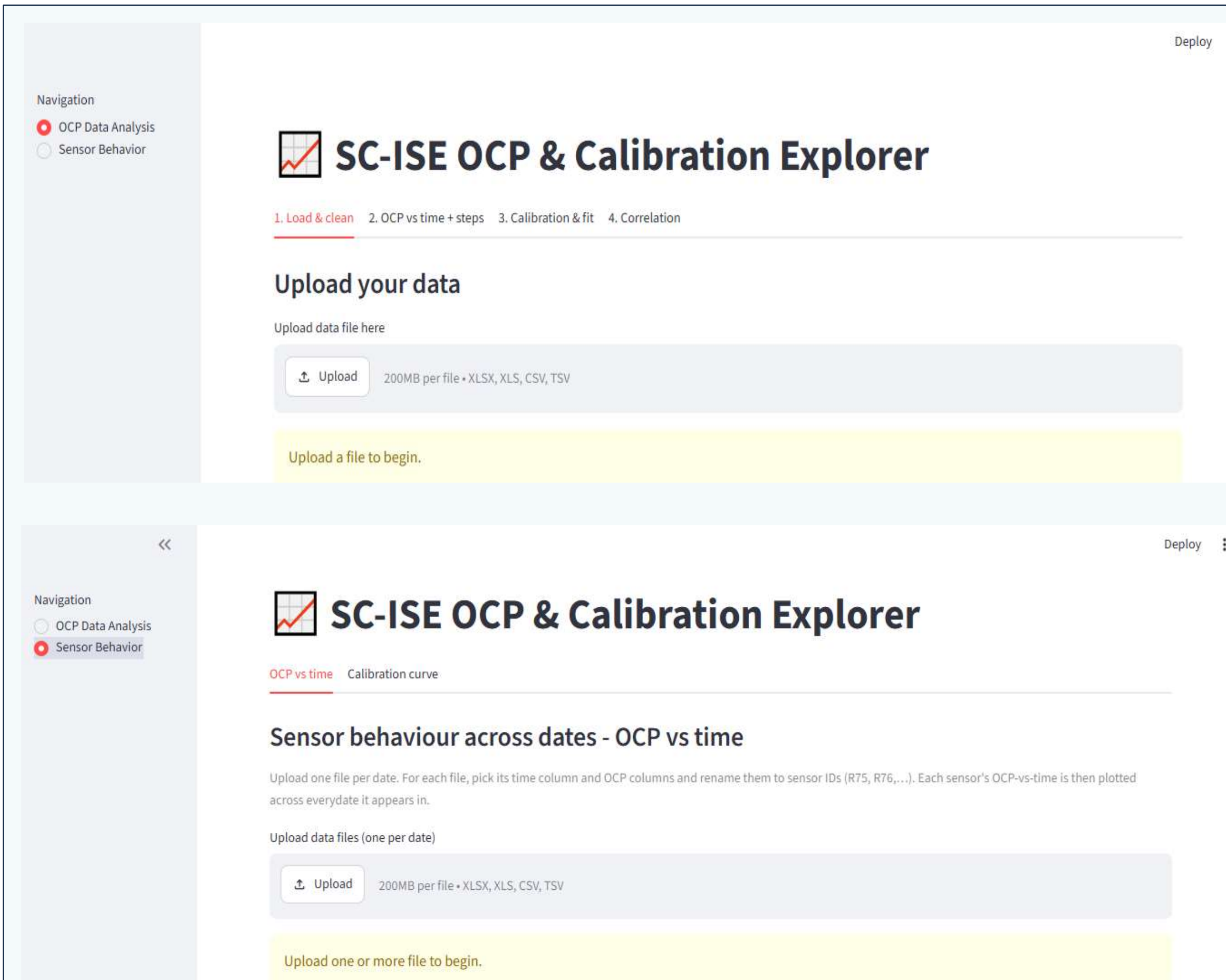
Left: raw multi-step response; the pipeline locates the steady-state window at each concentration and extracts one OCP. **Right:** the linear region and LOD are found automatically while points below the LOD are dropped from the fit.

4. Results — Automated Calibration of an 8-Electrode SC-ISE Array Across Five Sessions



Each panel: steady-state OCP vs. logC for one screen-printed electrode. **Filled markers** = automatically selected linear region; **Open markers** = excluded points. **Legends** report the fitted slope (mV/dec) and R² for each session — all extracted without manual curve-picking. Most electrodes show a consistent super-Nernstian response ($\approx 46\text{--}57\text{ mV/dec}$, $R^2 \geq 0.98$) for Calcium ions.

5. No-Code Web Interface



Built with Streamlit: labmates upload spreadsheets and instantly get calibration parameters, per-electrode plots, and QC flags — replacing per-user Jupyter notebooks with one shared, consistent tool.

6. Conclusions

- A single automated pipeline converts raw potentiometric data into calibration parameters (slope, intercept, R²) for every electrode with no manual curve-picking, giving reproducible, user-independent results.
- Reorganizing the data from date-wise to sensor-wise lets the pipeline track behavior of each electrode over time.
- Delivered as a no-code Streamlit app, the workflow analyzes a full multi-electrode dataset in seconds instead of hours — converting tedious manual analysis into time and energy the group can spend on research.
- This laid the foundation for ML driven analysis where we extract features from raw data and train a ML models on them.

7. Future Work

Concentration: Predict concentration by training a ML model on features extracted from the sensor response.

Electrode behavior clustering: Run clustering ML algorithms on the various parameters (slope, R², intercept) to group electrodes into stable / drifting / dead automatically.

Sensation limit: Train ML model to automatically detect sensation limit of sensor.

8. Acknowledgements

Supported by the NDnano Undergraduate Research Fellowship (NURF) and the Notre Dame Department of Chemical & Biomolecular Engineering. I thank Prof. Jennifer Schaefer and Prof. Nosang V. Myung for their mentorship and also PhD. Govinda Devkota and grad student Raúl Chávez for guiding me throughout this project and for providing the SC-ISE calibration data, .